Mr Dimelow

3.1 Engineering materials

3.1.1 Materials and their properties

Students should have knowledge and understanding of the following groups/classifications of engineering materials. Students should be able to identify these materials based on their physical appearances and the following properties:

- ***toughness/brittleness***
- ***ductility***
- ***malleability***
- ***hardness***
- ***strength and stiffness***

Students should also be able to demonstrate knowledge and understanding of the behavioural characteristics of each of these materials during handling/machining.

Materials and their Properties





In this presentation you will:

- explore the characteristics used to describe engineering materials

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Knowledge Requirements

The learner can:

State the properties associated with materials:

- *Ductility*
- *Malleability*
- *Hardness*
- *Toughness*
- *Strength*
- *Brittleness*
- *Plasticity*
- *Elasticity*

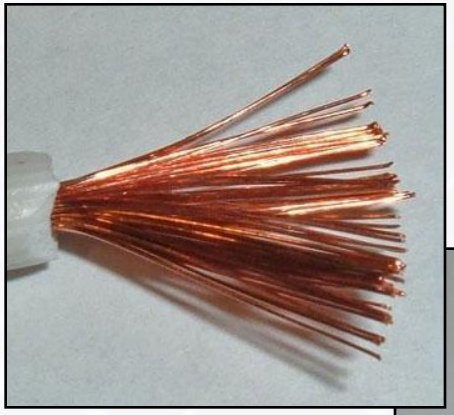
Introduction

There is a huge range of engineering materials in everyday use.

Materials are described and selected based on their characteristics and properties.









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Material Properties

Materials are selected for products because of their ***'properties'***. As an product designer it important to get this right. In order to make this choice you need to know the ***'Properties of the Material'***.

1. ***Does the material need to be light or heavy?***
2. ***How strong must it be?***
3. ***Does it need to be flexible or stiff?***
4. ***Is colour important?***
5. ***What kind of finish is required?***
6. ***Is cost very important?***

Once these and other questions have been answered, you will be able to choose a suitable material.

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Spot the Material!

1. Think about something that you use everyday, such as a bike, computer or even your toothbrush….

2. How many different materials are used in it? and how many of them can you identify?

3. Working in pairs make a list of at least **TEN** commonly used materials in engineering.

4. For each material that you have identified, list at least **TWO** important properties that make them useful in engineering.

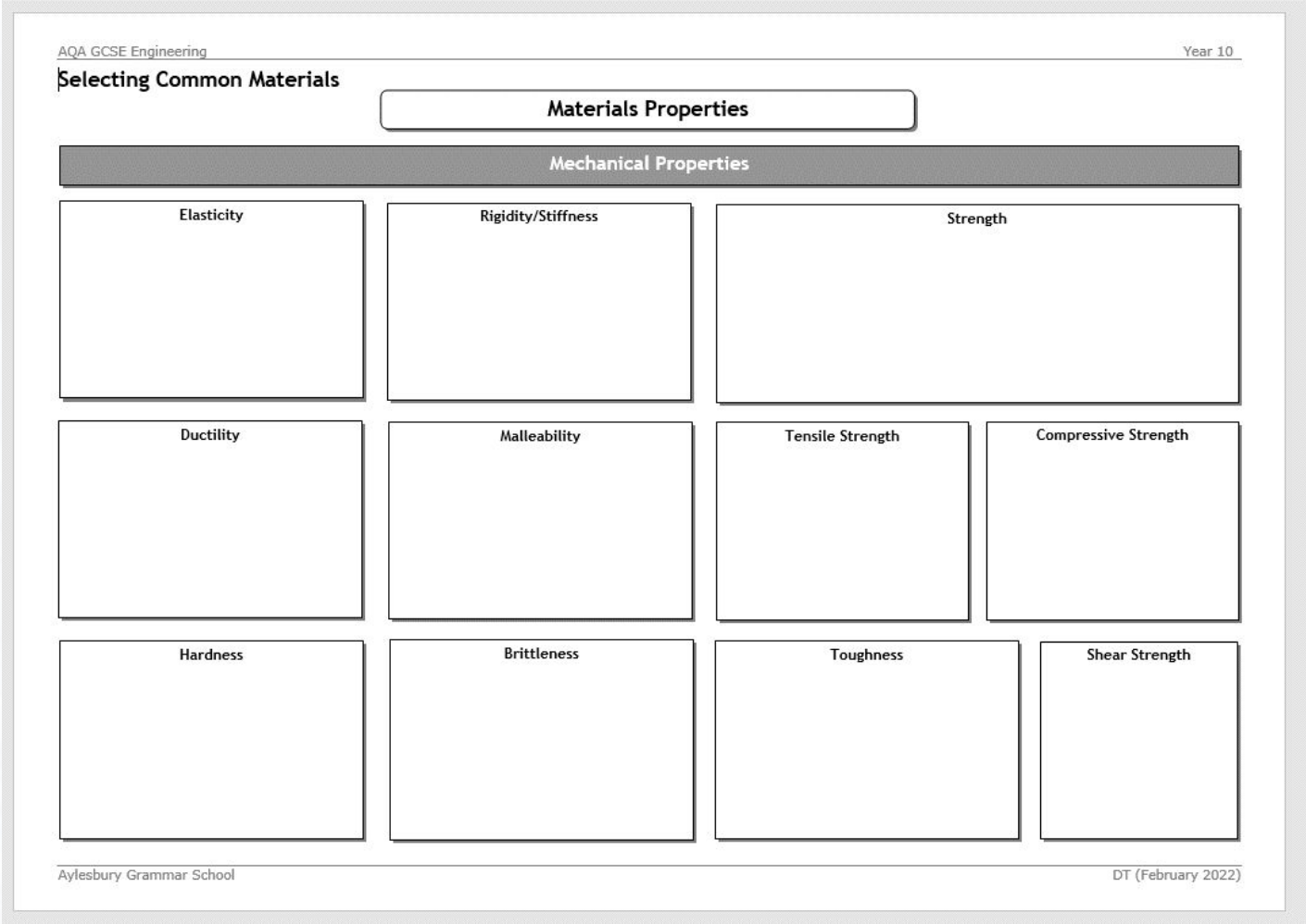
Next >

Material Properties

We are now going to look at the different ***properties*** of ***materials***…

Next >

Material Properties



AQA GCSE Engineering — Year 10

Selecting Common Materials

Materials Properties

Mechanical Properties

Elasticity	Rigidity/Stiffness	Strength	
Ductility	Malleability	Tensile Strength	Compressive Strength
Hardness	Brittleness	Toughness	Shear Strength

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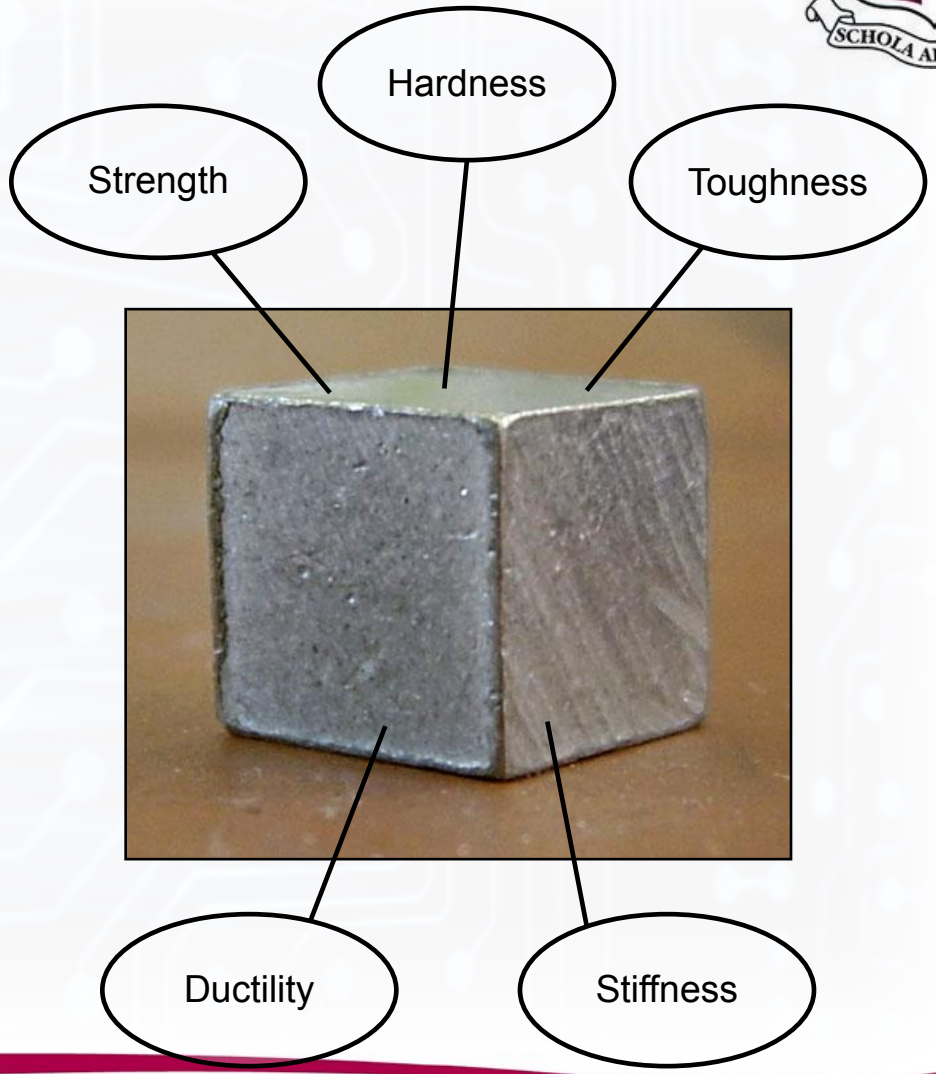
Complete the sheet by filling in the *definitions of the 'Material Properties'. Include* ***diagrams*** where possible.

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Mechanical Properties

Mechanical properties describe how a material responds to the ***application of a force***.

They include ***tensile strength, compressive strength, hardness, toughness ductility***...



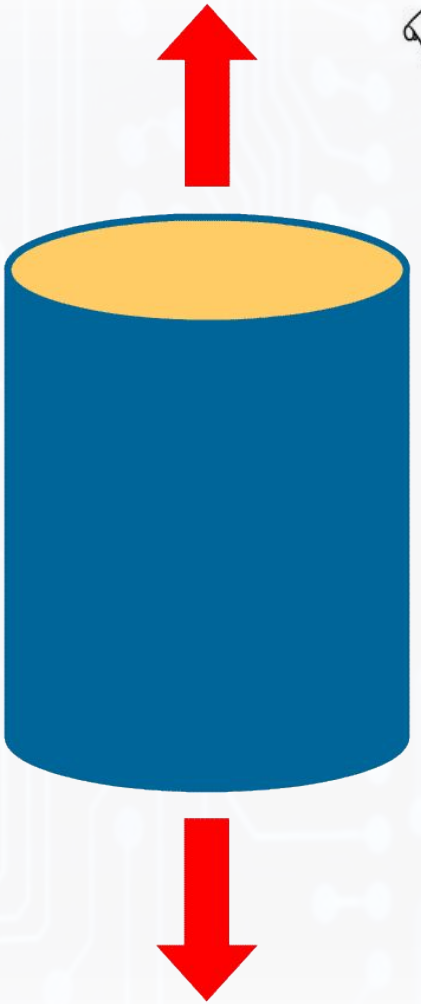
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Elastic Deformation

When a material undergoes ***stress***, it ***changes shape***.
This is called ***deformation***.

Sometimes deformation is ***reversible***.

If the material ***returns*** to its ***original shape*** when the *force is removed*, this is called ***elastic deformation***.

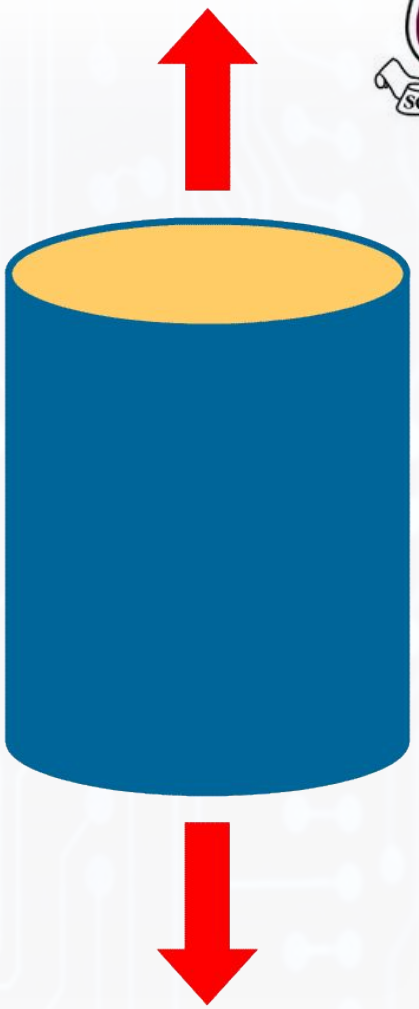


Effect of ***'heat'*** on the property – ***'Decreases Elasticity'***

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Plastic Deformation

In some materials, when the ***force is removed***, the material ***does not return*** to its ***original shape***.



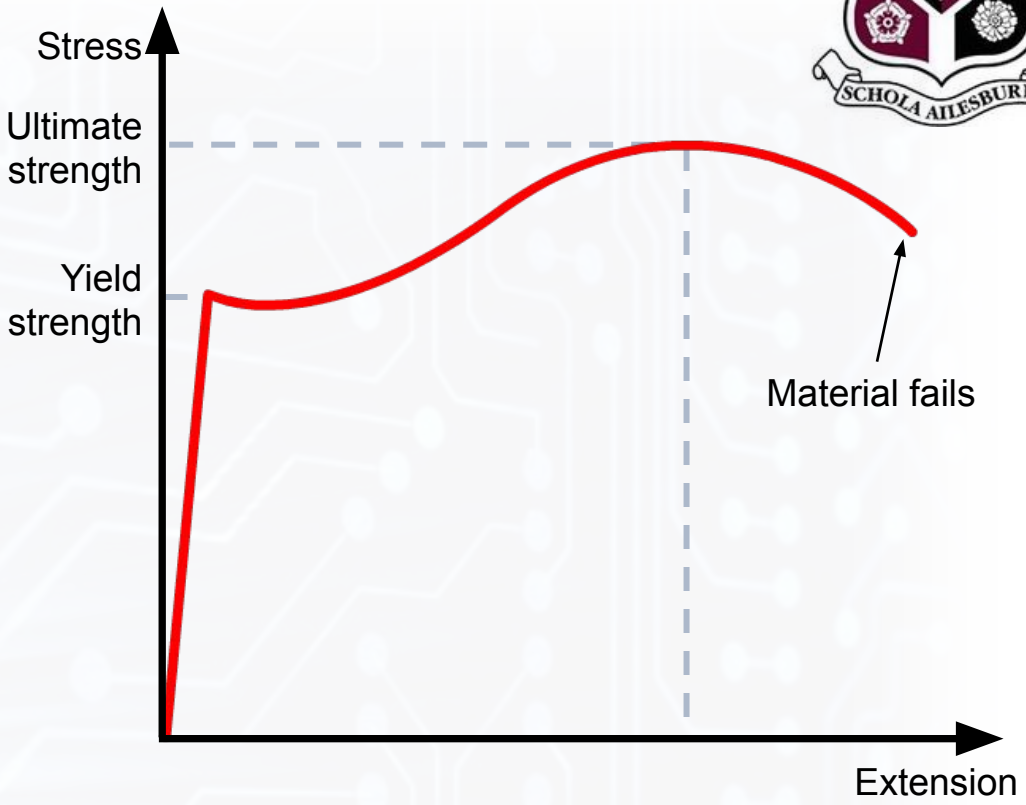
This type of ***deformation*** is ***permanent***.

This is called ***plastic deformation***.

Effect of ***'heat'*** on the property – ***'Increases Plasticity'***

Strength

The ***maximum stress*** that a material can withstand is called its ***strength***.



The ***yield strength*** is the ***stress*** at which ***plastic deformation*** starts to occur.

The ***ultimate strength*** is the ***maximum stress*** that a material can withstand ***before it fails*** .

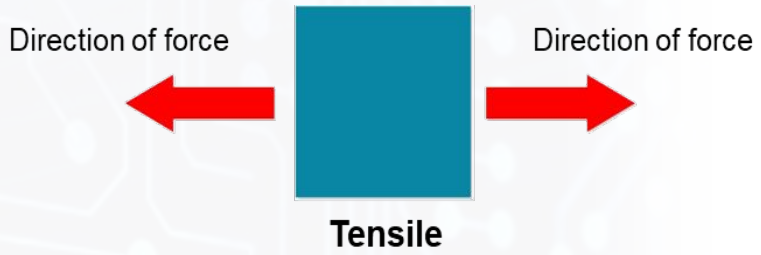
Effect of ***'heat'*** on the property – ***'Decreases Strength'***

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Types of Strength

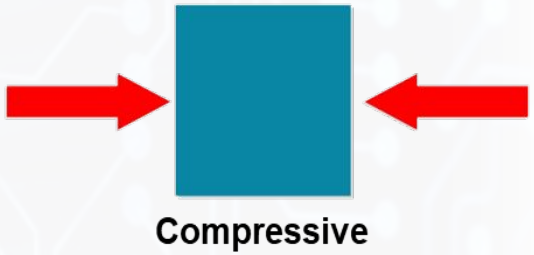
TENSILE STRENGTH

This is the ability of a material to withstand ***tensile (stretching) loads*** without rupture occurring. The material is in ***tension***.



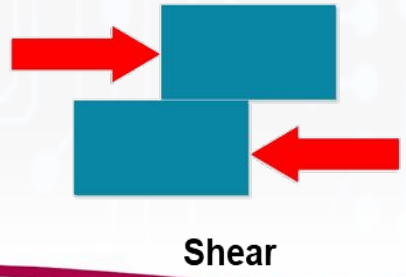
COMPRESSIVE STRENGTH

This is the ability of a material to withstand ***compressive (squeezing) loads*** without being crushed or broken. The materials is in ***compression***.



SHEAR STRENGTH

This is the ability of a material to withstand ***offset loads*** without rupture occurring.



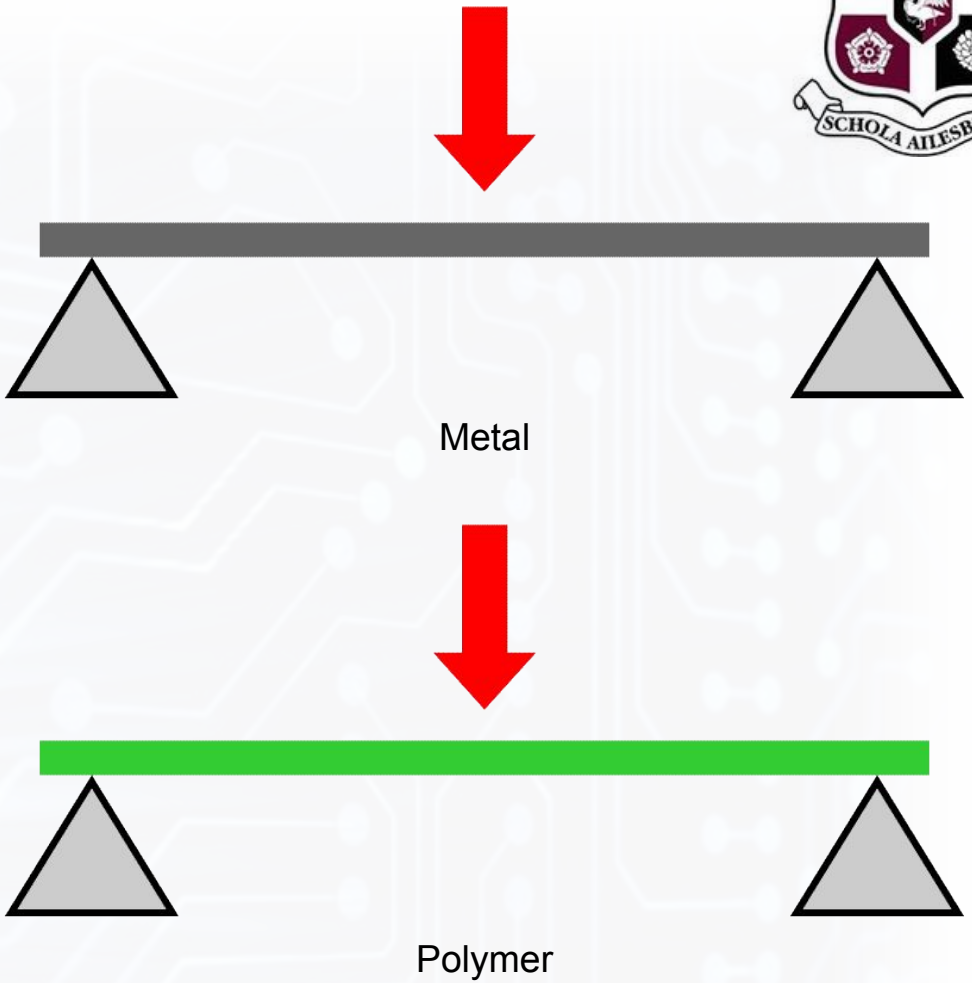
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Stiffness

One of the most important engineering properties of materials is ***stiffness***.

Stiffness relates to how ***rigid*** or how ***flexible*** a material is.

It describes the material's ***resistance to deformation*** when a force is applied.



Effect of ***'heat'*** on the property – ***'Decreases Stiffness'***

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Ductility

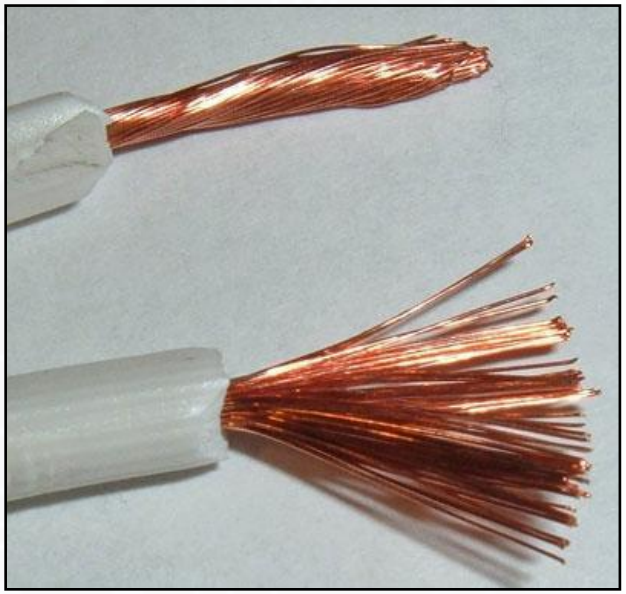
In manufacturing, materials are *deformed* into the desired shape.

A material's ability to be ***plastically deformed*** into a range of shapes is known as ***ductility***.

A ***ductile material***, such as ***copper***, can be ***drawn out*** into a long thin wire.







Effect of ***'heat'*** on the property – ***'Increases Ductility'***

Next >

Malleability

In manufacturing, materials are ***deformed*** into the desired shape.

A material's ability to be ***plastically deformed*** into a range of shapes is known as ***malleability***.

A ***malleable material***, such as aluminium, can be ***beaten*** or ***rolled out*** into a thin sheets, forging steel components, planishing a copper bowl.







Effect of ***'heat'*** on the property – ***'Increases Malleability'***

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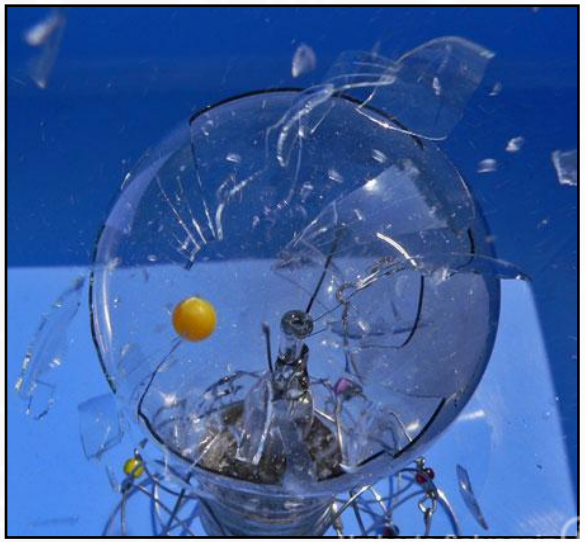
Toughness

Toughness is the ability of a material to ***absorb energy without failing***.



A ***tough material***, such as steel, can ***absorb a great deal of energy*** *before fracturing.*

Brittle materials may have ***high strength*** , but they are ***not tough*** .

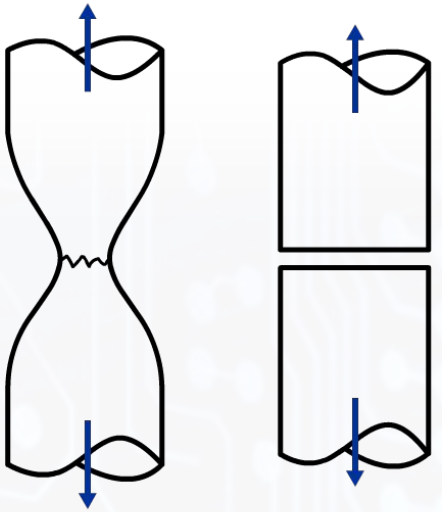


A ***tough material*** has a mixture of ***strength*** and ***ductility***.

Effect of ***'heat'*** on the property – ***'Decreases Toughness'***

Brittle

When a material ***under stress breaks without deforming***, it is described as ***brittle***.



Brittle materials do not undergo any ***deformation*** before failure.



Ceramics and glass are well known examples of ***brittle materials***.

Effect of ***'heat'*** on the property – ***'Decreases Brittleness'***

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Hardness

In engineering, ***hardness*** is the *resistance of a material to local deformation*.

Hard materials, such as glass and ceramics, are ***resistant to scratching, cutting or indentation***.

Soft materials, such as plastics, copper and gold, ***can be cut or scratched easily***.



Effect of ***'heat'*** on the property – ***'Decreases Hardness'***

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Simple Workshop Tests

For testing the ***properties*** of ***materials***…

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Practical Task 1:

Go into the workshop and conduct simple tests of mechanical properties for a range of engineering materials.

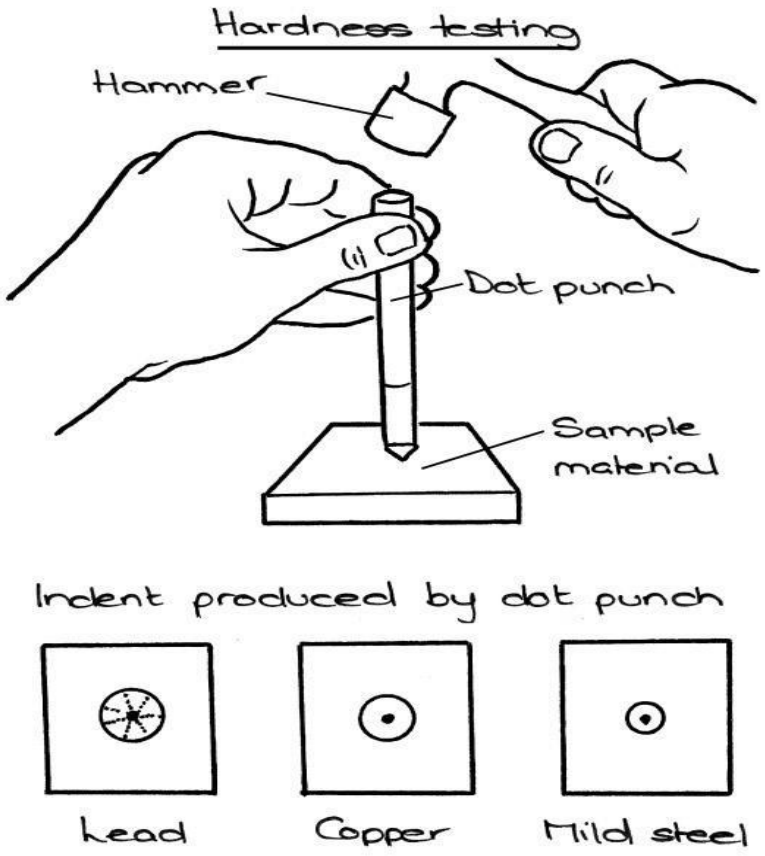
Hardness Test

Use a *hammer* & *centre/dot punch* to test the ***hardness*** of the materials listed below:

- ***Acrylic***
- ***Aluminium***
- ***Brass***
- ***Mild Steel***

Use the same force blow when testing each material.

Place the materials in order of hardness. Record your results.



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Practical Task 2:

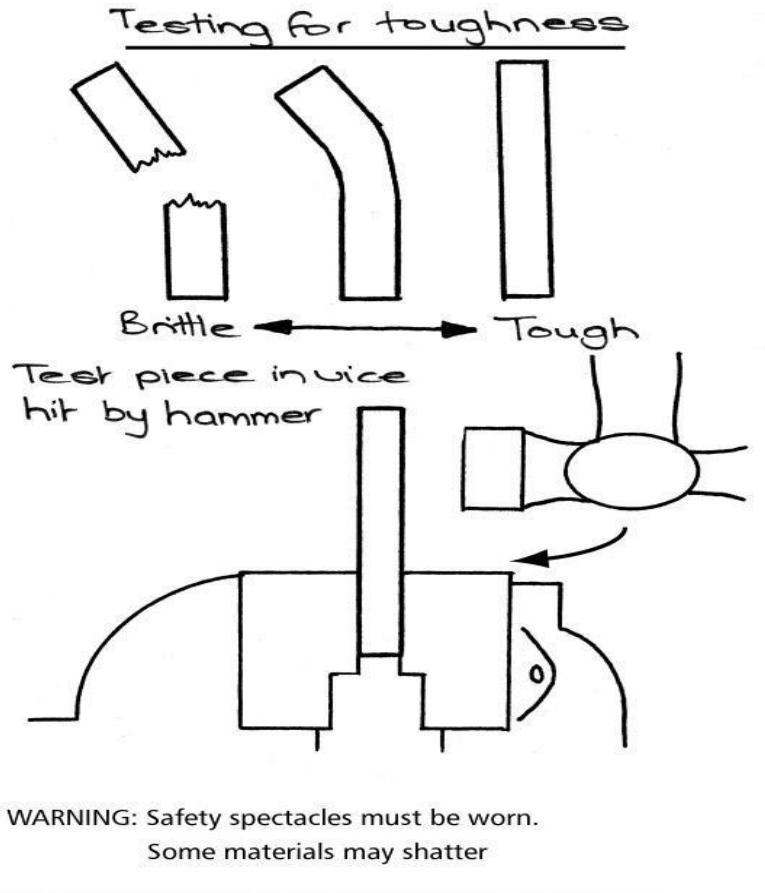
Toughness Test

Using a ***bench vice*** and an ***engineer's hammer*** test the ***toughness*** of the materials listed below:

- ***Acrylic***
- ***Aluminium***
- ***Mild Steel***

Fix one end of a *test piece* in a *bench vice* then *strike* ***'once'*** with an *'engineer's hammer'*.

Compare the materials by repeating the test on samples of equal sizes. Record your results.



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Practical Task 3:

Ductility/Malleability Test

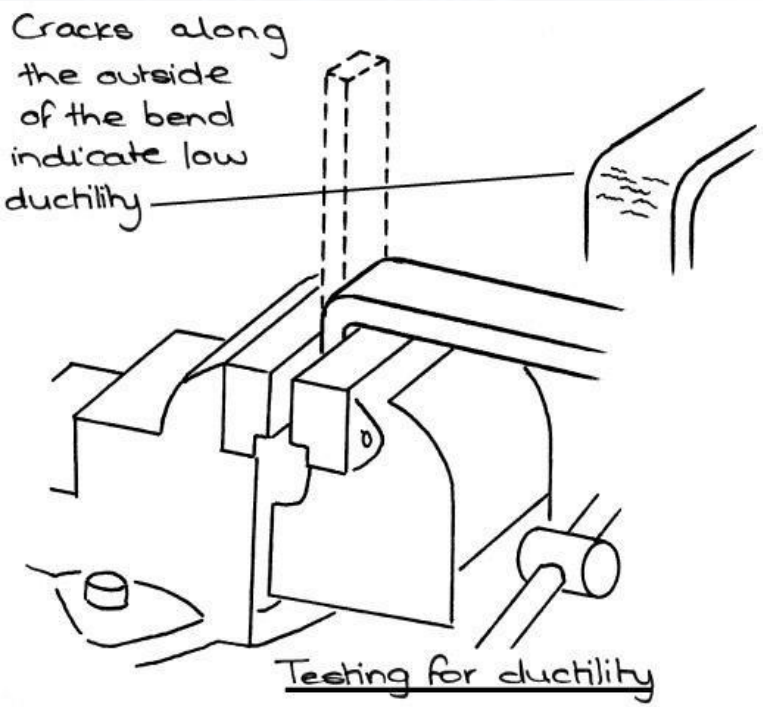
Using a ***bench vice*** and an ***engineer's hammer*** test the ***ductilty*** of the materials listed below:

- ***Acrylic***
- ***Aluminium***
- ***Mild Steel***

Fix one end of a ***'cold'*** *test piece* in a *bench vice* then ***strike*** with an *'engineer's hammer' until the material* ***'bends'*** *(or fractures if it is* ***not*** *ductile)*.

A ***ductile*** *metal* will ***not*** show ***any cracks***.

Mild steel *may* have ***some cracking***.



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Practical Task 4:

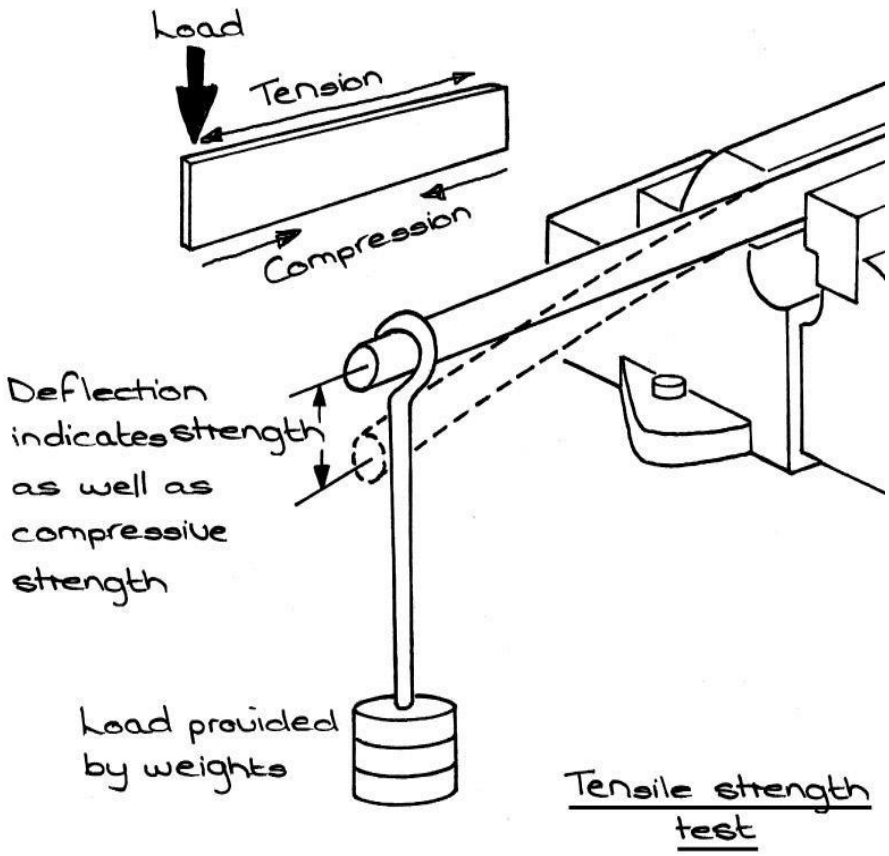
Tensile Strength Test

Using a ***bench vice*** and some ***masses*** test the ***tensile strength*** of a mild steel bar.

Fix one end of a *test piece* in a *bench vice and safely attach masses to the end.*

Measure the ***deflection*** of the mild steel bar.

Try testing ***other metals*** of the same size. ***Compare*** your results.



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Summary

In this presentation you have seen:

- the main characteristics that are used to describe engineering materials
- definitions of strength, stiffness, ductility, toughness and hardness

End